

CS221: End Sem Paper 1 30 Mins

21 Questions 30 Minutes

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Points: 33/45

✓ **Correct** 2/2 Points

1

Given two 5-bit numbers A and B, how many combinations are there where $A = B$

32

✗ **Incorrect** 0/2 Points

2

Which of the following must always be true if $(A \text{ XOR } B \text{ XOR } C \text{ XOR } D) = 0$?
(more than one options are correct – marks will only be given when all correct options are selected)

☐ $(A \text{ XOR } C) = (B \text{ XOR } C)$ ✓

☐ $A \text{ OR } B \text{ OR } C \text{ OR } D = 0$

☐ $(A' \text{ XOR } C') = (B' \text{ XOR } C')$ ✓

☐ $A \text{ AND } B \text{ AND } C \text{ AND } D = 0$

✓ **Correct** 2/2 Points

3

In number system with base=10 and 2, if $(29.25)_{10} = (X)_2$, what is

- ☐ X = 10111.10
- ☐ X = 11101.10
- ☒ X = 11101.01 ✓
- ☐ X = 10111.01

✓ **Correct** 2/2 Points

4

Choose the correct choice

- (i) An irredundant expression is not necessarily minimal
- (ii) The minimal expression always unique

- ☐ (i) True, (ii) True
- ☐ (i) False, (ii) False
- ☒ (i) True, (ii) False ✓
- ☐ (i) False, (ii) True

✓ **Correct** 2/2 Points

5

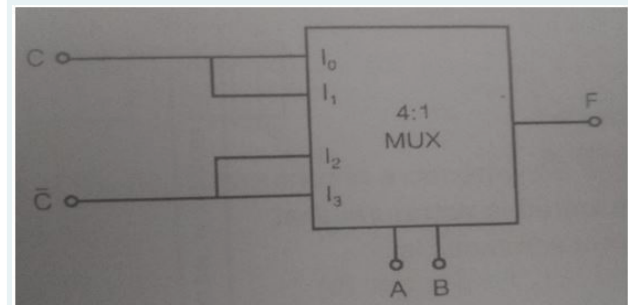
A 5-bit Binary number 10011 will be encoded in Hamming code as follows:

- ☐ 001100111
- ☐ 101100101
- ☐ 011000111
- ☒ 101100111 ✓

✓ **Correct** 2/2 Points

6

Choose the correct option for the function represented by the MUX



- ☒ A xor C ✓
- ☐ B xor C
- ☐ B xnor C
- ☐ A xnor C

✗ **Incorrect** 0/3 Points

7

Consider the given expression and with reference to it's cube-literal matrix .Answer the given question.

Expression: $uwz + uxz + vwz + vxz + yz + uv$

Let, 'n' denote the number of prime rectangles we obtain from it's cube literal matrix and 'm' be the unique co-kernels obtained. Fill in the blank with the sum of 'n' and 'm' .

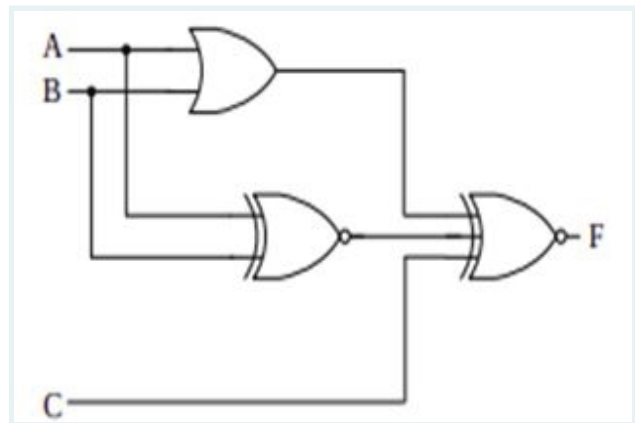
9

Correct answers: 11

✓ **Correct** 2/2 Points

8

For the output F to be 1 in the logic circuit in the question, the input combination should be

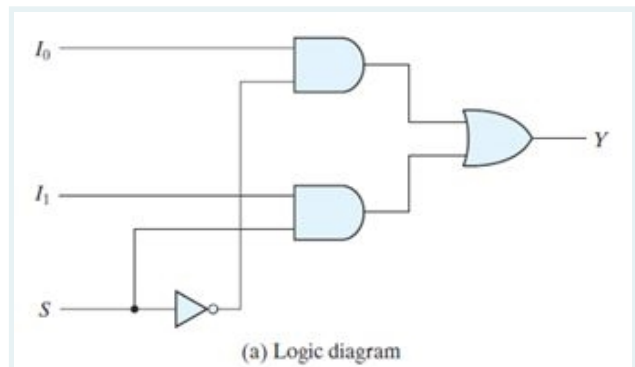


- ☒ A = 0, B = 0, C = 1 ✓
- ☐ A = 0, B = 1, C = 0
- ☐ A = 1, B = 0, C = 0
- ☐ A = 1, B = 1, C = 1

✓ **Correct** 2/2 Points

9

Consider the following logic diagram of 2 to 1 MUX. If we want to realize $XOR(A, B)$ using this multiplexer, what would be the value you should set for I_0 , I_1 and S in terms of A , B , A' , B' , 0 and 1?



- ☒ $I_1 = B', S = A, I_0 = B$ ✓
- ☐ $I_1 = 0, S = B, I_0 = A$
- ☐ $I_1 = A, S = , I_0 = 0$
- ☐ $I_1 = 1, S = B, I_0 = A'$

✓ **Correct** 2/2 Points

10

Number of AND gates (assuming 6 input AND gates) required to implement a decoder with 5 inputs and 1 enable line are ____.

32

✓ **Correct** 2/2 Points

11

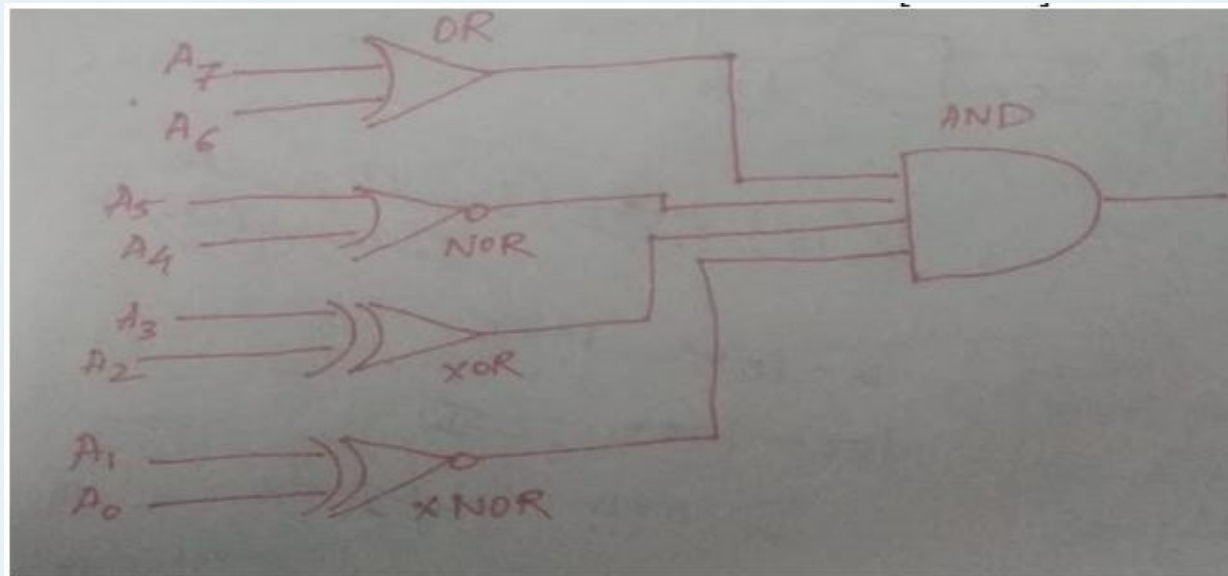
A 6-bit odd parity bit generator has to be made without any optimization. How many AND gates are required for achieving the circuit? (Write the Integer value only)

32

✓ **Correct** 3/3 Points

12

A device is enabled using 8 address bits (A7, A6,, A1, A0) as shown below. Calculate the number of addresses for which this device is enabled



12

✓ **Correct** 2/2 Points

13

Which of the following can be a kernel for an expression? (More than one option may be correct. No partial marking)

☒ $xy + wz + y$ ✓

☒ $y + wx$ ✓

☐ $xy + xz + x$
☐ wx
✓ **Correct** 2/2 Points

14

Total number of kernels for the expression $pu + qu + pv + rv + pst + qst + rst$ is (Consider that the corresponding co-kernel should have at least 1 literal)

6

✓ **Correct** 2/2 Points

15

Any Boolean function can be implemented using a _____

- ☐ Decoder with an AND gate
- ☐ Encoder with an AND gate
- ☐ Encoder with an AND gate
- ☒ Decoder with an OR gate ✓

✓ **Correct** 2/2 Points

16

The most optimal and correct representation of the following expression is
A. $(A + B) + B.(A' + B)$

- ☒ $A + B$ ✓
- ☐ $AA + B$
- ☐ $A + AB$
- ☐ $A' + AB$

✗ **Incorrect** 0/3 Points

17

Use the Quine-McCluskey method and list all the essential prime implicants of the below function F?

$$F(A, B, C, D, E) = \Sigma(3, 6, 7, 8, 10, 14, 15, 17, 20, 24, 25, 28)$$

- ☐ $A'B'DE, AC'D'E, ACD'E'$

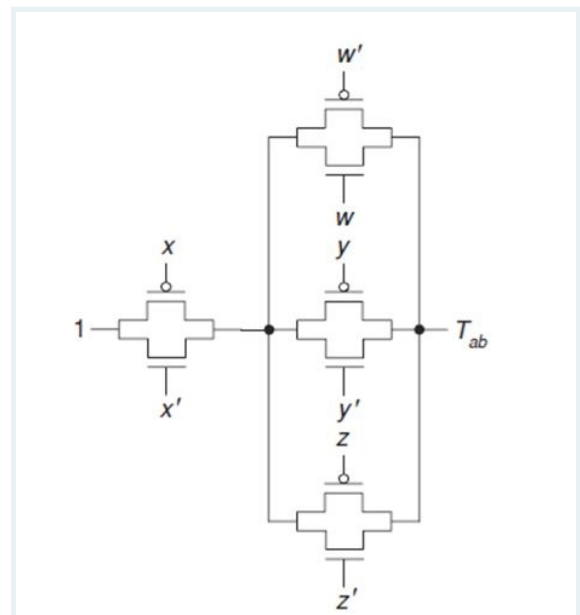
- ☐ $A'B'DE, ACD'E', AC'D'E, A'CD$ ✓
- ☐ $A'B'CD, AC'D'E, ACD'E'$
- ☐ $A'B'CE, AC'D'E, A'CD$

✓ **Correct** 2/2 Points

18

Consider the following CMOS design. What will be the output when

(i) $w=0, x=0, y=1, z=1$ and (ii) $w=0, x=1, y=0, z=0$



- ☐ (i) 1 and (ii) 0
- ☐ (i) 0 and (ii) 1
- ☐ (i) 1 and (ii) 1
- ☒ (i) 0 and (ii) 0 ✓

✗ **Incorrect** 0/2 Points

19

Determine essential prime implicants of the following function in sum-of-products form. How many essential prime implicants are there?

$$f(a, b, c, d) = \Sigma(1, 3, 7, 9, 10, 11, 13, 15)$$

- ☐ 4 ✓
- ☐ 3

☐ 5☐ 2

✓ **Correct** 2/2 Points

20

What would be the possible values of m and n, if 73_m in base-m number system is base-n number system?

☐ 9, 13☒ 8, 11 ✓☐ 8, 16☐ 10, 12

✗ **Incorrect** 0/2 Points

21

If $X = -45$ and $Y = -31$, where X and Y are 8-bit numbers in 2's complement form, the result $X - Y$ in 8-bit 2's complement form is

☒ 00001110☐ 11110010 ✓☐ 11101110☐ 11100010

out your password.

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